

Scalability & Future Plan

Date	11 July 2026
Team ID	
Project Name	Rising waters - Flood Prediction using Machine Learning
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses historical rainfall dataset only	Prediction accuracy may decrease for changing climate patterns	High
2	Supports prediction only through manual user input	Not suitable for automated large-scale monitoring	Medium
3	No real-time weather API integration	Cannot provide live flood prediction updates	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited simultaneous users	Deploy on cloud with load balancing to handle more users
2	Data Storage	Local dataset and model files	Use cloud database and scalable storage services
3	Performance	Local Flask application	Optimize model inference and deploy on high-performance servers
4	Security	Basic input validation	Add authentication, HTTPS, and secure API access

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	Integrate real-time weather API	2 Months	Improves prediction accuracy with live weather data
Phase 2	Add district-wise flood risk visualization on maps	3 Months	Better decision-making for disaster management
Phase 3	Develop Android mobile application	5 Months	Enables users to access predictions from anywhere
Phase 4	Implement SMS and email alert notifications	6 Months	Provides instant flood warnings to users and authorities